MMBench Is Your Multi-modal Model an All-around Player (2307.06281v5)

Abstract

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> Traditional benchmarks like VQAv2 or COCO Caption provide quantitative performance measurements but lack fine-grained ability assessment and robust evaluation metrics

Lack of robust evaluation metrics and fine-grained ability assessment.

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> MMBench is meticulously curated with well-designed quality control schemes, surpassing existing similar benchmarks in terms of the number and variety of evaluation questions and abilities

Good quality control while maintaining numbers.

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> MMBench introduces a rigorous CircularEval strategy and incorporates large language models to convert free-form predictions into pre-defined choices, which helps to yield accurate evaluation results for models with limited instruction-following capabilitie

Open ended predictions --> pre-defined choices ==> CircularEval strategy.

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> MMBench incorporates multiple-choice questions in both English and Chinese versions, enabling an apples-to-apples comparison of VLMs' performance under a bilingual context

Bilingual benchmark, one-to-one comparison for exploring language biases.

Conclusion

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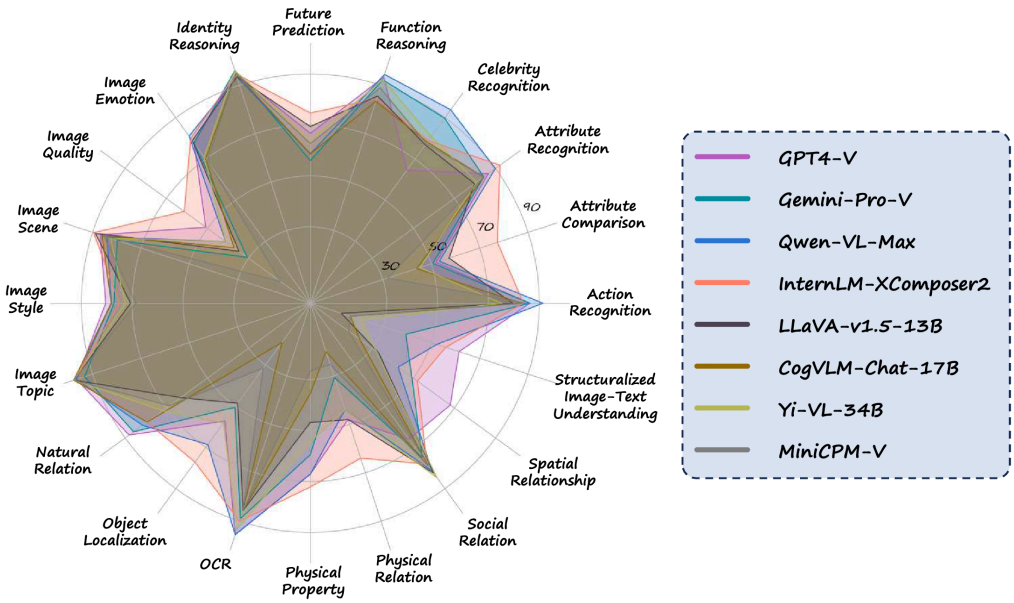
> Considering the limited instruction following ability of some VLMs, to yield more accurate evaluation results, we additionally adopt LLMs to extract choices from the model's predictions

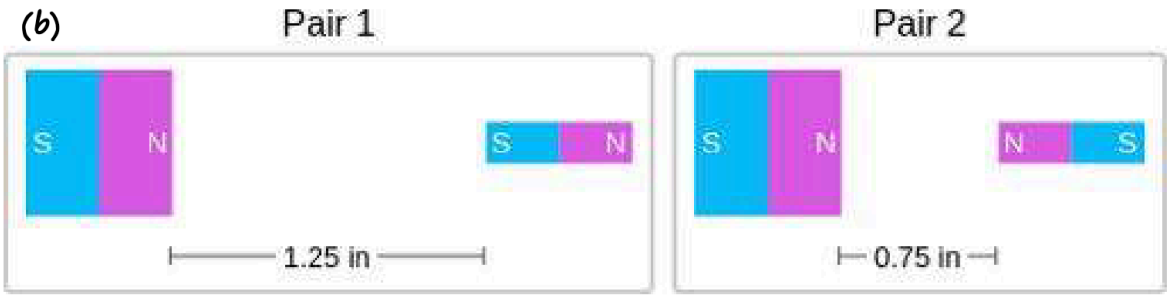
Use LLM-as-a-Judge.
One important observations:

1. Authors use different architectures and param sizes ==> different model families and params.

Tables and Figures



Overall results, models seem to perform worse on Image quality ability.



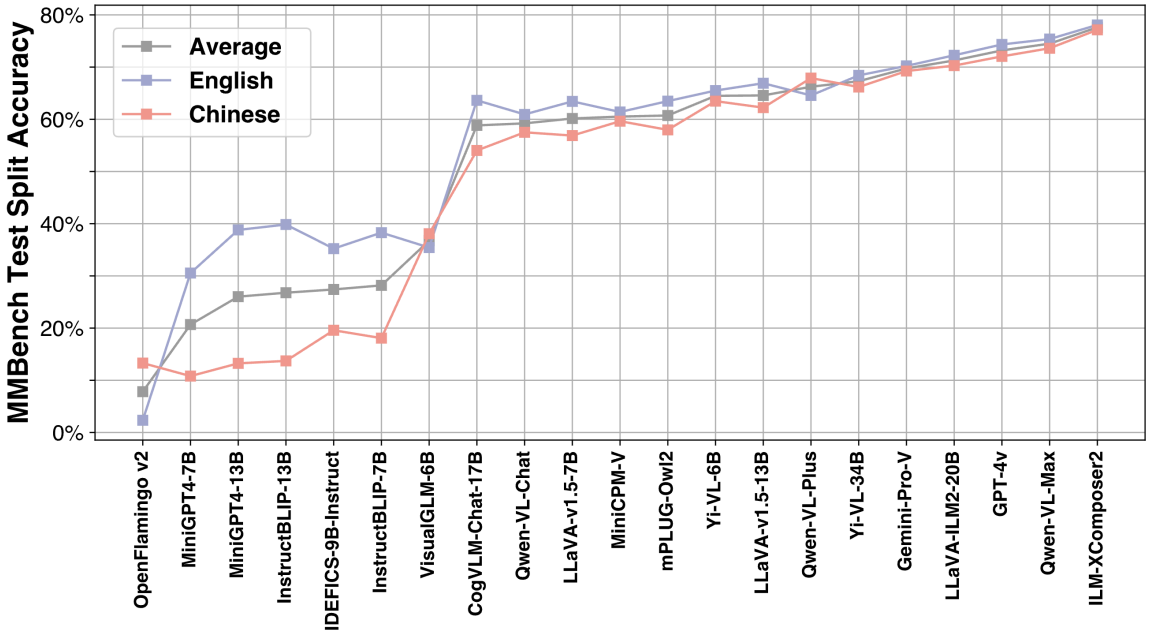
English Version (Original)

QUESTION. Think about the magnetic force between the magnets in each pair. Which of the following statements is true?
A. The magnitude of the magnetic force is smaller in Pair 2.
B. The magnitude of the magnetic force is smaller in Pair 1.
C. The magnitude of the magnetic force is the same in both pairs.

Chinese Version (Translated)

QUESTION. 考虑每对磁铁之间的磁力。以下哪个陈述是正确的?
A. 第二对磁铁之间的磁力大小较小。
B. 第一对磁铁之间的磁力大小较小。
C. 两对磁铁之间的磁力大小相同。

Bilingual question for one-to-one correspondence. Difficulty of questions can be questionable. Human verification might need expert users.



Some models perform poorly in Chinese language. Majority perform well in both English and Chinese.

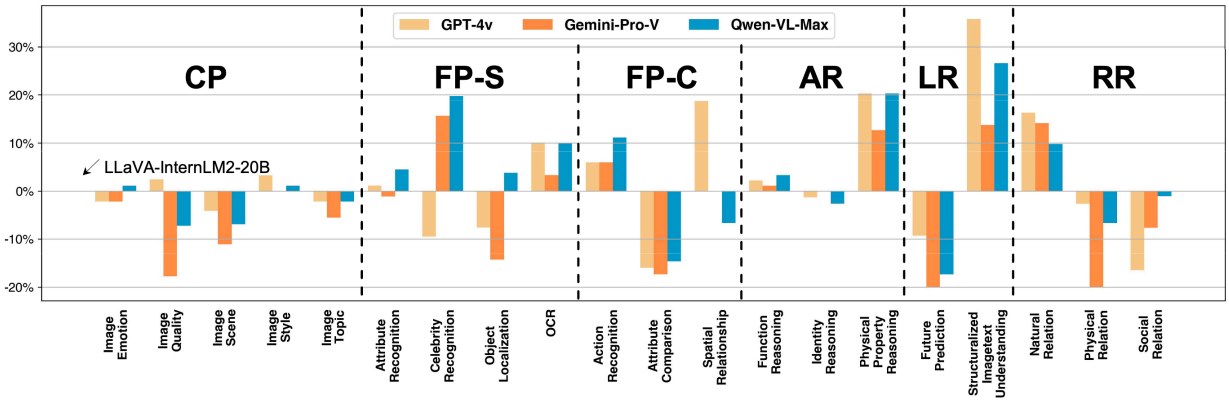


Figure 9: **Proprietary VLMs *vs.* Open-Source ones at a fine-grained level.**

How are scores negative?

Introduction

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> The lack of quantitative assessment poses a considerable challenge for comparing various models

Motivation: Lack of a quantitative benchmark.

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> 1. False Negative Issues: Most existing evaluation metrics require an exact match between the prediction and the reference target, leading to potential limitations. For instance, in the VQA task, even if the prediction is “bicycle” while the reference answer is “bike”, the existing metric would assign a negative score to the prediction, resulting in a considerable number of false-negative samples

Authors highlight the issue of false negatives - leading to potential category pollution. This point is taken care of in our work.

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> Given the aforementioned challenges, recent studies, such as OwlEval [49] and LVLM-eHub [48] propose human-involved subjective evaluation strategies, aiming to address existing methods’ limitations by incorporating human judgment

> and perception in the evaluation process. OwlEval artificially constructs 82 open-ended questions based on images from public datasets and employs human annotators to assess the quality of VLM predictions. Similarly, inspired by FastChat [53], LVLM-eHub develops an online platform where two models are prompted to answer the same question related to an image. A participant then compares the answers provided by two models. Subjective evaluation strategies offer numerous benefits. These include accurate matching, where humans can precisely correlate a prediction with the target, even when expressed in different words, and comprehensive assessment, where humans are inclined to juxtapose two predictions considering multiple facets. The ultimate score is computed as the mean score across diverse abilities, facilitating a holistic evaluation of the model's capabilities

Some works also employ human annotation and judgement, but it is not scalable, introduces human bias. Human evals require small dataset sizes, which can affect statistical stability.

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> employ GPT-4 to match a model's prediction to candidates choices in a multi-choice question and then output the label for the matched choice.

LLM as a judge, judge = GPT-4

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> We conduct a comparison between GPT-4-based choice matching and human evaluations, and discovered that GPT-4 can accurately match human assessments in 91.5% of cases, demonstrating its good alignment and robustness as a choice extractor

Human verification of the judge.

Dataset construction

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> n pursuit of genuine general artificial intelligence (AGI), vision-language models (VLMs) are also expected to exhibit strong perception and reasoning abilities.

> Therefore, we adopt Perception and Reasoning as level-1 (L-1) abilities in our taxonomy. After that, we incorporate more fine-grained ability dimensions into the taxonomy, and categorize them into six L-2 and twenty L-3 ability dimensions. We display the ability taxonomy in Figure 2 and you can find detailed definitions of each fine-grained ability in the Appendix.

Hierarchical arrangement of taxonomies - allowing for fine-grained analysis of each ability.

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> Raw data collected from volunteers may include wrong or unqualified samples. During investigation, we find that there exist two major patterns for such samples: i) the answer to the question can be inferred with text-only inputs, which makes it inappropriate for evaluating the multimodal understanding capability of VLMs; ii) the sample is simply wrong, either with a flawed question, choices, or an incorrect answer. We design two strategies to filter those low-quality samples, which is visualized in Figure 3(a). We adopt 'majority voting' to detect text-only samples: data samples are inferred with state-of-the-art LLMs (GPT-4 [37], Gemini-Pro [44], etc.). If more than half of the LLMs can answer the question correctly with text-only inputs, the question will be manually verified and then removed if it is unqualified. To detect wrong samples, we also implement an automatic filtering mechanism.

There are 2 major cases when questions can be wrong:

1. Question can be answered with text-only --> If more than half of the VLMs can answer the question with text-only, then the question is discarded.
2. Question is simply wrong --> If all VLMs fail to answer a question, it is rejected.
3. NOTE: Before rejection, each rejected question is manually verified.

Evaluation

Circular Eval

Question is fed to the model multiple times and checks if a VLM succeeds in all attempts.

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> Under this setting, each question is fed to a VLM N times (N is the number of choices). Each time, circular shifting is applied to the choices and the answer to generate a new prompt for VLMs (example in Figure 6). A VLM is considered successful in solving a question only if it correctly predicts the answer in all circular passes. In practice, once a VLM fails on a circular passes, there is no need to infer the remaining passes, which makes the actual cost of CircularEval less than N × under practical scenarios. CircularEval can achieve a good trade-off between robustness and cos

CircularEval prevents model biases for certain choice to creep into the eval.

Extracting VLM prediction

Use Judge model to extract prediction.

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> While there exist VLMs that perfectly follow the multiple-choice format and achieve high success rates (> 99%) in heuristic matching, all proprietary models and a significant proportion of opensource VLMs failed to generate well-formatted outputs.

Can be worthwhile to do a similar sanity check for our project for this claim.

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> Figure 5 reports the alignment rates (extracted choices are exactly the same) between LLMs and humans. We find that a great number of LLMs can complete the task well and achieve decent alignment rate with human. Among proprietary LLMs, GPT-4 achieves the highest level of alignment rate, which is 91.5%, while GPT-3.5-Turbo and Qwen-Max achieve around 85%

Human alignment of the judge responses.

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> CircularEval is such a challenging setting that it even makes state-of-the-art proprietary VLMs (GPT-4v, Qwen-VL-Max

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etc.) suffer from ~10% Top-1 accuracy drops. In the following experiments, we adopt the more rigorous and well-defined CircularEval as our default evaluation paradigm

Circular eval can cause performance drop in SOTA VLMs.

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> LLM plays a vital role. From the evaluation results, we find that the large language model (LLM) adopted plays a vital role in the VLM performance. For instance, all LLaVA series VLMs (v1.5-7B, v1.5-13B, InternLM2-20B) adopt the same vision backbone and are trained with the same multimodal corpus, while switching the LLM from Vicuna-v1.5 [53] to the more powerful InternLM2-20B [45] leads to steady improvement across all L-2 capabilities (especially significant for reasoning tasks). The scaling also holds for variants with different sizes from the same LLM family. By adopting the 13B variant of Vicuna rather than the 7B variant, VLMs in the MiniGPT, InstructBLIP, and LLaVA v1.5 series outperform their 7B counterparts by 8.3%, 1.5%, and 3.5% overall Top-1 accuracies on the MMBench-test split, respectively.

Backbone plays an important role --> Is this relevant?

Fine-grained analysis

Table 3: **CircularEval results on MMBench `test` set (L-2 abilities).** Abbreviations adopted: LR for Logical Reasoning; AR for Attribute Reasoning; RR for Relation Reasoning; FP-C for Fine-grained Perception (Cross Instance); FP-S for Fine-grained Perception (Single Instance); CP for Coarse Perception. Models are sorted by the ascending order of overall accuracy (intra-group). Open-source models tagged with * incorporate in-house data in model training.

Model	Overall	CP	FP-S	FP-C	AR	LR	RR
Large Language Models							
GPT-4-Turbo (0125) [37]	2.9%	0.6%	1.2%	4.1%	3.7%	4.9%	7.4%
OpenSource VLMs							
OpenFlamingo v2 [4]	2.3%	1.1%	3.5%	1.5%	5.3%	0.0%	2.7%
MiniGPT4-7B [56]	30.5%	37.0%	31.8%	17.2%	49.8%	9.2%	25.6%
IDEFICS-9B-Instruct [26]	35.2%	48.3%	31.3%	29.6%	47.8%	11.4%	25.2%
VisualGLM-6B [13]	35.4%	40.2%	38.5%	26.2%	47.8%	19.6%	29.5%
InstructBLIP-7B [11]	38.3%	46.7%	39.0%	31.8%	55.5%	8.7%	31.0%
MiniGPT4-13B [56]	38.8%	44.6%	42.9%	23.2%	64.9%	8.2%	32.9%
InstructBLIP-13B [11]	39.8%	47.2%	42.9%	21.0%	60.4%	12.5%	38.8%
Qwen-VL-Chat* [6]	60.9%	68.5%	67.7%	50.2%	78.0%	37.0%	45.7%
MiniCPM-V [39]	61.4%	65.6%	69.4%	51.3%	70.6%	35.3%	59.7%
LLaVA-v1.5-7B [32]	63.4%	70.0%	68.0%	57.7%	77.6%	33.2%	56.2%
mPLUG-Owl2 [50]	63.5%	68.1%	69.1%	55.8%	78.4%	37.0%	57.0%
CogVLM-Chat-17B [47]	63.6%	72.8%	66.6%	55.4%	71.4%	33.7%	62.0%
Yi-VL-6B* [2]	65.5%	72.8%	72.9%	56.2%	75.5%	41.3%	55.4%
LLaVA-v1.5-13B [32]	66.9%	73.1%	72.4%	60.3%	75.5%	35.9%	65.5%
Yi-VL-34B* [2]	68.4%	72.0%	78.0%	54.7%	81.2%	38.6%	68.2%
LLaVA-InternLM2-20B [10]	72.3%	78.3%	76.6%	68.2%	78.4%	46.2%	69.4%
InternLM-XComposer2* [12]	78.1%	80.4%	83.5%	73.0%	83.7%	63.6%	74.4%
Proprietary VLMs							
Qwen-VL-Plus [6]	64.6%	66.5%	79.1%	50.2%	73.9%	42.9%	57.8%
Gemini-Pro-V [44]	70.2%	70.0%	78.9%	65.9%	82.9%	46.2%	65.9%
GPT-4v [37]	74.3%	77.6%	73.8%	71.5%	85.3%	63.6%	68.6%
Qwen-VL-Max [6]	75.4%	74.8%	87.2%	67.0%	85.3%	54.9%	70.5%

Performance results of VLMs (open-source and proprietary). GPT-4 Turbo is the "blind" baseline.

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> We observe that proprietary models significantly outperform the open-source ones under two major scenarios: i) Structuralized image-text understanding, which requires VLMs to understand complex codes, tables, diagrams, or layouts. ii) Tasks requiring external knowledge to solve, which correspond to abilities including celebrity recognition, physical property reasoning, natural relation reasoning, etc

When do prop. VLMs perform better than open source ^^